Original Article

Safety of acupuncture in oncology: A systematic review and meta-analysis of randomized controlled trials

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BACKGROUND: Acupuncture is frequently used to treat the side effects of cancer treatment, but the safety of this intervention remains uncertain. The current meta-analysis was conducted to assess the safety of acupuncture in oncological patients. **METHODS:** The PubMed, Cochrane Central Register of Controlled Trials, and Scopus databases were searched from their inception to August 7, 2020. Randomized controlled trials in oncological patients comparing invasive acupuncture with sham acupuncture, treatment as usual (TAU), or any other active control were eligible. Two reviewers independently extracted data on study characteristics and adverse events (AEs). Risk of bias was assessed using the Cochrane Risk of Bias Tool. **RESULTS:** Of 4590 screened articles, 65 were included in the analyses. The authors observed that acupuncture was not associated an with increased risk of intervention-related AEs, nonserious AEs, serious AEs, or dropout because of AEs compared with sham acupuncture and an active control. Compared with TAU, acupuncture was not associated with an increased risk of intervention-related AEs, serious AEs, or drop out because of AEs but was associated with an increased risk for nonserious AEs (odds ratio, 3.94; 95% confidence interval, 1.16-13.35; $P = .03$). However, the increased risk of nonserious AEs compared with TAU was not robust against selection bias. The meta-analyses may have been biased because of the insufficient reporting of AEs in the original randomized controlled trials. **CONCLUSIONS:** The current review indicates that acupuncture is as safe as sham acupuncture and active controls in oncological patients. The authors recommend researchers heed the CONSORT (Consolidated Standards of Reporting Trials) safety and harm extension for reporting to capture the side effects and better investigate the risk profile of acupuncture in oncology. ***Cancer* 2022;128:2159-2173**.

LAY SUMMARY:
- According to this analysis, acupuncture is a safe therapy for the treatment of patients with cancer.
- Acupuncture seems to be safe compared with sham acupuncture and active controls.

KEYWORDS: acupuncture, adverse event, meta-analysis, neoplasms, randomized controlled trial, safety.

INTRODUCTION

Cancer incidence is rapidly growing worldwide, with an estimated 18.1 million new cancer cases in 2018.[1,2] Cancer and its treatment lead to impairments in health-related quality of life not only in patients during treatment but also in cancer survivors.[3-6] Safe and effective therapies are needed to improve patients' symptom burden.

According to the World Health Organization, traditional and complementary medicine can make a significant contribution to the goal of ensuring healthy lives and promoting well-being for patients at all ages.[7] An easy, accessible therapy that belongs to traditional and complementary medicine is acupuncture.[7] In the National Cancer Institute (NCI) *NCI Dictionary of Cancer Terms*, acupuncture is defined as *a technique of inserting thin needles through the skin at specific points on the body to control pain and other symptoms*.[8] Acupuncture, a therapy of Traditional Chinese Medicine, was first

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described in the Inner Canon of Yellow Emperor (Huang Di Nei Jing) in the second century before Christ.[9] In 2016, the Division of Cancer Treatment and Diagnosis, Office of Cancer Complementary and Alternative Medicine at the NCI held an expert panel on acupuncture for cancer symptom management.[10] The experts discussed the efficacy data from randomized controlled trials (RCTs) of acupuncture and reviewed safety issues as well.[10] In a large, population-based study with 97,733 patients, only minor intervention-related side effects were reported.[11] The authors concluded that acupuncture is a safe treatment with only minor side effects when applied by licensed practitioners.[11] In a prospective, observational study with 229,230 nononcological patients, 8.6% of patients reported at least 1 adverse effect because of acupuncture. Nevertheless, the expert panel was concerned about higher risks of infection and bleeding in patients on active oncological treatment during neutropenia and thrombocytopenia. *Adverse effects/side effects* suggest causality, whereas the term *adverse events* (AEs) describes harmful events that occur during a trial.[12] In 2.2% of patients, 1 of the reported AEs required treatment. In 2 patients, a pneumothorax was caused by acupuncture, with a number needed to harm of 2 in 250,000 patients[13]; 1 of those patients needed hospital treatment, and the other only needed observation.[14] In both observational studies, local pain from needling, bruising, minor bleeding, and orthostatic problems were the most common side effects.[10,11] One meta-analysis of prospective clinical trials on acupuncture-related AEs included 11 studies with a total of 845,637 patients.[15] In that analysis, the overall risk of at least 1 acupuncture-related AE during a series of acupuncture treatments was estimated to be 9.31 per 100 patients treated.

However, data on AEs for acupuncture in oncology are less well studied compared with acupuncture for chronic pain. In oncology, treatment-related comorbidities like neutropenia and thrombocytopenia and increased risks because of disease need to be taken into account.[10] In some systematic reviews, no data concerning intervention-related AEs of acupuncture treatment are reported[16-22] or they are not systematically analyzed.[23-29] Only a few meta-analyses[30-32] deal with adverse effects/AEs, and those are limited to patients who have breast cancer or a symptom like chemotherapy-induced peripheral neuropathy.[33] To examine the safety of acupuncture in cancer care, we systematically accessed and meta-analyzed the frequency of AEs of acupuncture in RCTs of patients with cancer and cancer survivors.

TABLE 1. Complete Search Strategy for PubMed

1	Acupuncture[MeSH] OR Acupuncture Therapy[MeSH]
2	acupunct*[tiab] OR acupress*[tiab] OR acupoints*[tiab] OR electroacupunct*[tiab] OR electro-acupunct*[tiab] OR auriculoacupunct*[tiab] OR ear-acupunct*[tiab]
3	1 OR 2
4	Neoplasms[MeSH]
5	Neoplasm*[tiab] OR cancer[tiab] OR tumor*[tiab] OR tumour*[tiab] OR malignan*[tiab] OR oncolog*[tiab] OR carcinoma*[tiab] OR leukaemia[tiab] OR leukemia[tiab] OR lymphoma*[tiab] OR sarcoma*[tiab]
6	4 OR 5
7	Randomized Controlled Trial[MeSH] OR Randomized Controlled Trial[pt] OR Controlled Clinical Trial[pt]
8	randomized[tiab] OR randomised[tiab] OR RCT[tiab] OR (random*[tiab] AND allocat*[tiab]) OR (random*[tiab] AND assign*[tiab]) OR placebo*[tiab] OR sham[tiab] OR group[tiab] OR trial[tiab]
9	7 OR 8
10	3 AND 6 AND 9

Abbreviation: MeSH, Medical Subject Heading; tiab, title/abstract (in PubMed).

MATERIALS AND METHODS

We followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines[34] and recommendations of the Cochrane Collaboration[35] in planning, conducting, and reporting this systematic review and meta-analysis. The protocol for this systematic review and meta-analysis was registered on PROSPERO (CRD42019125542) and is available in full at https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=125542 (accessed June 23, 2020).

Literature Search

The 3 electronic databases PubMed/MEDLINE, the Cochrane Library (Central), and Scopus were searched from their inception through April 7, 2020. The complete search strategy for PubMed is provided in Table 1. No language restrictions were applied. In addition, reference lists of identified original articles and reviews were searched. Pairs of 2 review authors (M.D.H. and A.B. or M.D.H. and P.V.) independently screened titles and abstracts and read potentially eligible articles in full to determine whether or not they met eligibility criteria. Disagreements were discussed with a third review author (H.C. or H.H.) until consensus was reached.

Eligibility criteria

To be included, studies had to fulfill the following criteria:

Types of studies

RCTs and randomized crossover trials were eligible. No language restrictions were applied.

Types of participants

Studies that included oncological patients of any age and any treatment status were eligible. No restrictions regarding health status or sociodemographic characteristics were applied.

Types of interventions

Studies that compared any kinds of invasive acupuncture (including permanent needles, ear acupuncture, electroacupuncture) with treatment as usual (TAU), no treatment/waitlist, any other active control, or sham acupuncture/noninvasive acupuncture (eg, laser acupuncture) were eligible. *Active control* and *TAU* do not refer to oncological therapy in this article.

Types of outcome measures

Studies were included if they provided sufficient information about the occurrence of AEs. In cases of missing data, all study authors who provided at least some data about AEs in the original publication were contacted by e-mail. The NCI defines an AE as *an unexpected medical problem that happens during treatment with a drug or other therapy*.[36] Four types of AEs were differentiated: 1) intervention-related, 2) nonserious, 3) serious, and 4) dropout because of AEs. AEs were considered intervention-related when the authors of the original publication indicated a causal relationship. AEs were categorized as serious if they resulted in: 1) death, 2) life-threatening situations, 3) hospitalization, 4) disability or permanent damage, 5) congenital anomaly/birth defect, or 6) the need for medical or surgical intervention to prevent outcomes; all other events were regarded as nonserious.[37] If an AE was given as a reason for dropping out, it was considered a dropout because of an AE.

Data extraction

Two authors (M.D.H. and A.B. or M.D.H. and P.V.) independently extracted data on design, participants, interventions, control interventions, and AEs using an a priori developed extraction sheet. If discrepancies occurred, they were discussed with a third reviewer (H.C. or H.H.) until consensus was achieved. If data were reported insufficiently, the authors of the original publication were contacted by e-mail or social media. The data were re-checked if they corresponded with the reported outcomes in the article. If there were discrepancies, the authors were contacted again. If they did not respond, the conflicting reported AEs were not included in the meta-analysis.

Risk of Bias

Two reviewers (M.D.H. and S.A. or M.D.H. and P.V.) independently accessed the risk of bias using the Cochrane Risk of Bias Tool.[35] Risk of bias was assessed on the following domains: selection bias, performance bias, detection bias, attrition bias, and reporting bias. If discrepancies occurred, they were discussed with a third reviewer (H.C. or H.H.) until consensus was achieved.

Data Analysis

The 4 types of AEs were analyzed separately. The control interventions were grouped as sham acupuncture (noninvasive acupuncture at acupoints or nonacupoints), TAU/waitlists, or active control (pharmacologic, psychological interventions, exercise, etc). The oncological treatment phases were grouped as: 1) active oncological treatment (chemotherapy, radiotherapy, palliative treatment), 2) survivorship (active oncological treatment, except for endocrine treatment completed), 3) surgery, 4) diverse (patients in various oncological treatment phases), and 5) best supportive care. Because there were insufficient data concerning follow-up, we deviated from the protocol and accessed only postmeasurement data. Regarding crossover trials, poor reporting impeded our attempts to include all data points,[38,39] so we treated them as parallel trials and included only data that were available before the crossover.

A standardized Excel sheet was developed to calculate odds ratios, which were determined by dividing the odds of an AE in the intervention group (the number of participants in the intervention group experiencing this type of AE divided by the number without this AE) by the odds of an AE in the control group. A value of 0.5 was added to each cell if the authors reported zero AEs in at least 1 of the intervention groups.[40,41] The log(odds ratios) and respective standard errors were calculated. Review Manager 5 software, version 5.3 (The Nordic Cochrane Center) was used for the meta-analysis, using random-effects models and the generic inverse-variance method.

Heterogeneity

Heterogeneity was assessed using I^2 statistics, which were categorized as *low* (0%-24%), *moderate* (25%-49%), *substantial* (50%-74%), or *considerable* (75%-100%) heterogeneity.[35] A significance level of $P \leq .10$ was applied because of the low power of this test regarding small sample sizes and because only a few studies were included.

Subgroup and Sensitivity Analyses

For significant outcomes, subgroup analyses were performed for the subgroups *type of cancer* (breast cancer, other cancer) and *intervention type* (acupuncture, ear acupuncture, electroacupuncture) if the outcomes were significant in the main analysis. Supplementary to the protocol, we also analyzed *oncological treatment phase* subgroups (active oncological treatment, surgery, survivorship, best supportive care, and diverse). Sensitivity analyses for studies with a low risk of selection bias were conducted to test the robustness of the results.[42]

Publication Bias

Publication bias was assessed by visual analysis of funnel plots if ≥10 studies were included.[35] Two reviewers (M.D.H. and P.V.) independently inspected the funnel plots. Roughly symmetrical funnel plots were regarded as an indication of a low risk of publication bias, whereas asymmetrical funnel plots were regarded as an indication of a high risk of publication bias.[35]

RESULTS

Literature Search

The electronic search yielded 4590 results. After removing duplicates, 3074 records were screened and, of those, 2877 articles were excluded after screening of titles and abstracts. Of 197 screened full texts, 94 publications were preliminarily included and were checked for safety-related data (Fig. 1).[1-134] Of those, 65 of 94 publications (69.1%) were included in the qualitative analysis because they reported at least 1 type of AE,[43-107] 15 (16%) did not report any type of safety-related data,[108-122] and 14 (14.9%) had insufficient or contradictory data.[118,123-135] Only 14 studies (14.9%) reported data on all 4 categories of AEs. Three studies concerning crossover trials reported insufficient data for inclusion in the meta-analyses.[58,65,92] Finally, 62 studies were included in the meta-analyses.[43-57,59-64,66-91,93-106,136] Of those, 29 RCTs were included in meta-analyses for acupuncture versus sham control,[45,46,48,49,52,53,55,57,59,66,68,70,72,75,77-80,82,88-90,94,96,97,99,102,103,107] 25 RCTs were included in meta-analyses for acupuncture versus TAU/waitlist control,[46,47,49-51,54,56,62,64,67,69,71,73,76,81,83-85,91,95,98,101,104-106] and 9 RCTs were included in meta-analyses for acupuncture versus active control[43,60,61,63,74,86,87,93,100] (Table 2).

Study Characteristics

The characteristics of the included studies are listed in Table 3,[43-107] which provides an overview of country of origin, sample size, type of cancer, treatment phase, and types of experimental and control interventions. The 65 studies included 4943 participants, with sample sizes ranging from 10 to 302 (mean, 74.9 participants). One study included pediatric patients,[65] and all other studies included adult participants. The mean age of participants ranged from 13.6 to 67.7 years (mean, 54.5 years). The percentage of female participants ranged from 0% to 100% (mean, 81%). Only 19 studies reported ethnicity. In those studies, the percentage of Caucasian participants ranged from 25.5% to 100% (mean, 79.4%).

Risk of Bias

Supporting Table 1 shows the detailed risk-of-bias assessment for the 65 studies. Thirty studies (46%) had a low selection bias, zero studies had a low performance bias, 30 studies (46%) had a low detection bias, 54 studies (83%) had a low attrition bias, 24 studies (40%) had a low reporting bias, and 55 studies (85%) had a low other bias.

Analyses of Adverse Events

An overview of the included studies is provided in Table 3. In meta-analyses comparing acupuncture versus sham acupuncture and acupuncture with active control, no differences in the frequency of intervention-related AEs, nonseries AEs, serious AEs, and dropout because of AEs were found (Table 2). We also observed no differences in the frequency of AEs comparing acupuncture with TAU for intervention-related AEs, serious AEs, and dropout because of AEs (Table 2 and Figs. 2-4).[44,46,47,49-51,54,56,62,64,67,69,71,73,76,79-81,83,85,88,91,94-96,98,101,104-106] In the comparison of acupuncture versus TAU, more nonserious AEs occurred in the acupuncture group than in the TAU group (odds ratio, 3.94; 95% confidence interval [CI], 1.16-13.35; $P = .03$) (see Table 3 and Fig. 5).[44,49,50,54,56,83]

Heterogeneity was low to moderate in all primary meta-analyses except for the comparison between acupuncture and active controls on intervention-related AEs ($I^2 = 72\%$; $\chi^2 = 25.27$; $P = .00$) (Table 2). When only studies with a low risk of selection bias were included, the heterogeneity (I^2) dropped to 44% and was no longer significant ($P = .11$). Subgroup analyses demonstrated that the heterogeneity could be explained by subgroup differences because cancer survivors had a particularly high level of heterogeneity ($I^2 = 74\%$; $P = .002$). For the same comparison on intervention AEs, the subgroups acupuncture ($I^2 = 82\%$; $P = .0007$) and electroacupuncture ($I^2 = 85\%$; $P = .009$) had comparably high levels of heterogeneity.

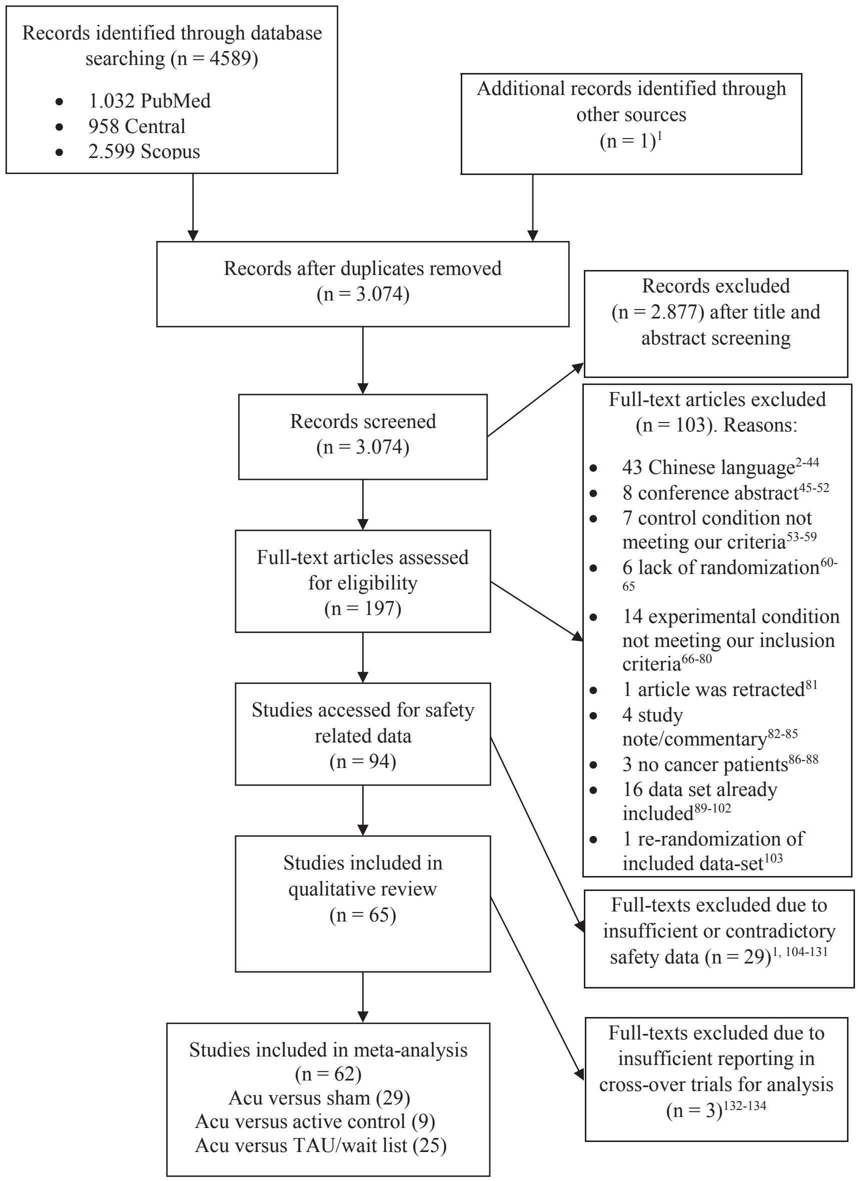

FIGURE 1. This is a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram of the current study (relevant references are indicated by superscript numbers). Acu indicates acupuncture; TAU, treatment as usual.

The total frequency of AEs in the acupuncture group was relatively low: 86 of 1278 patients (6.7%) experienced intervention-related AEs, 151 of 746 (20.2%) experienced nonserious AEs, 6 of 829 (0.7%) experienced serious AEs, and 31 of 1825 (1.7%) dropped out because of AEs.

Subgroup analyses indicated that there were no significant differences between groups concerning treatment phase, cancer diagnosis, or acupuncture type. However, for different types of cancer, more nonserious AEs were reported among patients who had breast cancer than for other cancer diagnoses compared with TAU (see Supporting Fig. 1).

Sensitivity analyses produced results similar to those of the primary analyses, except for nonserious AEs in the comparison of acupuncture versus TAU and of acupuncture versus active control (see Supporting Table 2). When only studies with a low risk of selection bias were included, the comparison between nonserious AEs because of acupuncture versus TAU was no longer significant (odds ratio, 2.42; 95% CI, 0.88-6.65; $P = .09$) (see Supporting Fig. 2), however, only 1 study was included

TABLE 2. Adverse Events of Acupuncture Versus Controls

Comparison	Outcome: Type of AEs	No. of Studies	No. of Events/Participants: Acupuncture	Control	Odds Ratio (95% CI)	*P* for Overall Effect	Heterogeneity: I^2, %	χ^2 Statistic [*P*]
Acupuncture vs sham: 29 trials	Nonserious AE	9	92/277	41/218	1.66 (0.99-2.81)	.06	0.0	6.51 [.59]
	Serious AE	12	1/383	1/327	0.92 (0.31-2.77)	.88	0.0	1.07 [1.0]
	Intervention-related AE	19	22/461	8/447	1.74 (0.88-3.42)	.11	0.0	5.10 [1.0]
	Dropouts because of AE	24	12/660	12/641	1.05 (0.56-1.97)	.88	0.0	5.32 [1.0]
Acupuncture vs TAU: 25 trials	Nonserious AE	6	45/189	10/206	3.94 (1.16-13.35)	.03	29.0	7.09 [.21]
	Serious AE	8	5/244	1/262	1.76 (0.51-6.11)	.37	0.0	1.73 [.97]
	Intervention-related AE	20	38/653	2/651	2.09 (0.94-4.66)	.07	0.0	12.41 [.87]
	Dropouts because of AE	27	19/893	13/901	1.46 (0.82-2.61)	.20	0.0	4.25 [1.0]
Acupuncture vs active control: 9 trials	Nonserious AE	6	14/280	35/448	0.49 (0.16-1.46)	.20	42.0	8.62 [.13]
	Serious AE	6	0/202	3/233	0.66 (0.15-2.90)	.58	0.0	0.91 [.97]
	Intervention-related AE	8	26/242	41/290	1.00 (0.26-3.92)	1.00	72.0	25.27 [.00]
	Dropouts because of AE	9	0/272	12/318	0.44 (0.15-1.29)	.14	0.0	1.66 [.99]

Abbreviations: AE, adverse event; CI, confidence interval; TAU, treatment as usual.

in the analysis. In the sensitivity analyses of nonserious AEs, the comparison between acupuncture and active control became significant, with more AEs in the control group (odds ratio, 0.30; 95% CI, 0.11-0.81; $P = .02$) (see Supporting Fig. 3).

Publication Bias

The funnel plots of the meta-analyses that included ≥10 studies were rated as generally symmetrical (see Supporting Figs. 4-8).

DISCUSSION

In this review, we strived to answer the question whether acupuncture is a safe treatment option for oncological patients. We found that acupuncture was not associated with an increased risk of intervention-related, nonserious, or serious AEs or of dropout because of AEs compared with sham acupuncture and different active controls, such as acupressure, active relaxation, or hydroelectric baths. Acupuncture was associated with an increased risk for nonserious AEs compared with TAU, but it was not associated with an increased risk of intervention-related or serious AEs or of dropout because of AEs. Sensitivity analyses showed that this effect might be caused by low study quality because the effect was not robust against selection bias. Furthermore, sensitivity analyses indicated that, when only studies with a low risk of selection bias were included, there was a higher frequency of nonserious AEs in the active control group compared with the acupuncture group. Subgroup analyses showed that, compared with TAU, patients with breast cancer who were receiving acupuncture were more prone to nonserious AEs than those with other cancers. However, this difference between groups was not statistically significant. In total, the frequency of AEs because of acupuncture was low (intervention-related AEs, 6.7%; nonserious AEs, 20.2%; serious AEs, 0.7%; dropout because of AEs, 1.7%). Substantial heterogeneity among our comparisons was only observed in the comparison between acupuncture and active controls, which may have been caused by very heterogeneous control interventions. Although we controlled for methodological bias, there was a large body of studies with insufficient reporting of AEs: Only 69.1% reported safety-related data adequate to be included in our analyses, and only 14.9% reported data concerning all 4 types of AEs.

Regarding the poor reporting of AEs, Ioannidis et al published a harm and safety extension of the CONSORT (Consolidated Standards of Reporting Trials) statement in 2004[12] to improve proper reporting in clinical trials. Nevertheless, a reflection of reporting standards by Hodkinson et al in 2013[137] showed that the reporting of harms is still poor in RCTs, with one-half of the included studies not adhering to the CONSORT checklist. An analysis in 2019[138] showed that only 29% of the studies included details of specific attempts to ascertain data on AEs and often only reported *patients with at least 1 event*, with 84% of studies ignoring repeated AEs. Covering the safety of clinical cancer trials in 2014, Sivendran et al observed

TABLE 3. Characteristics of the Included Studies

Reference	Origin	Sample size	Condition	Treatment Phase	Intervention (Type of Acupuncture)	Control Intervention(s)
Alimi 2003[43]	France	90	Diverse cancer	Survivorship	Ear acupuncture (permanent needles)	Active control (auricular acupressure at placebo points)
Arai 2013[44]	Japan	26	Colon cancer	Surgery	Ear acupuncture	TAU
Balk 2009[45]	USA	27	Breast cancer, 26 patients; endometrial cancer, 1 patient	AOT	Electroacupuncture	Sham acupuncture at acupuncture points
Bao 2013[46]	USA	51	Breast cancer	Survivorship (100% endocrine)	Acupuncture	Sham acupuncture at placebo points
Bao 2018[47]	USA	82	Breast cancer	Survivorship	Acupuncture	TAU
Beith 2012[48]	Australia	32	Breast cancer	AOT	Electroacupuncture	Sham electroacupuncture at acupuncture points
Bokmand & Flyger 2013[49]	Denmark	94	Breast cancer	Survivorship	Acupuncture	1) Sham acupuncture at placebo points; 2) TAU
Brinkhaus 2019[50]	Germany	150	Breast cancer	Chemotherapy	Acupuncture	TAU
Chae 2016[51]	Korea	10	Gastric cancer	Surgery	Electroacupuncture	TAU
Chen 2013[52]	China	60	Pancreatic cancer	Best supportive care	Electroacupuncture	Sham acupuncture at acupuncture points
Cheng 2017[53]	China	28	Lung cancer	Survivorship	Acupuncture	Sham acupuncture at acupuncture points
Crew 2007[54]	USA	21	Breast cancer	Survivorship	Acupuncture and ear acupuncture	TAU
Crew 2010[55]	USA	43	Breast cancer	Survivorship	Acupuncture and ear acupuncture	Sham acupuncture at placebo points
D'Alessandro 2022[56]	Brazil	29	Diverse cancer	Survivorship	acupuncture	TAU
Deng 2007[58a]	USA	72	Breast cancer	Survivorship	Acupuncture	Sham acupuncture at placebo points
Deng 2018[57] (2020 same data set)	USA	63	Multiple myeloma	Chemotherapy	Acupuncture	Sham acupuncture at placebo points
Enblom 2011[59]	Sweden	237	Diverse cancer	AOT	Acupuncture	Sham acupuncture at placebo points
Feng 2011[60]	China	80	Diverse cancer	Survivorship	Acupuncture	Active control (Fluoxetine)
Frisk 2008[61]	Sweden	45	Breast cancer	Survivorship	Electroacupuncture	Active control (hormone therapy with estrogen/progestogen)
Garcia 2008[62]	USA	78	Diverse cancer	Surgery	Electroacupuncture	TAU
Garland 2019[63]	USA	160	Diverse cancer	Survivorship	Acupuncture	Active control (CBT-I)
Giron 2016[64]	Brazil	50	Breast cancer	Chemotherapy/radiation	Acupuncture	TAU
Gottschling 2008[65a]	Germany	23	Diverse cancer (pediatric)	AOT	Acupuncture	TAU
Greenlee 2016[66]	USA	63	Breast cancer	AOT	Electroacupuncture	Sham acupuncture at placebo points
Han 2017[67]	China	104	Multiple myeloma	AOT	Acupuncture	TAU
Hershman 2018[68]	USA	110	Breast cancer	Survivorship	Acupuncture	Sham acupuncture at placebo points)
Jeon 2015[70]	Korea	14	Thyroid cancer	AOT	Acupuncture	Sham (Kim's sham acupuncture needles)
Jeon 2019[69]	Korea	26	Thyroid cancer	Survivorship	Acupuncture	TAU
Jung 2017[71]	Korea	40	Gastric cancer	Surgery	Electroacupuncture	TAU
Kim & Lee 2018[72]	Korea	30	Advanced diverse cancer	Palliative care	Permanent needle	Sham acupuncture at acupuncture points
Kong 2020[73]	Korea	25	Lung cancer	AOT	Acupuncture	TAU
Lesi 2016[74]	Italy	190	Breast cancer	Survivorship	Acupuncture	Active control (enhanced self-care)
Liljegren 2012[75]	Sweden	84	Breast cancer	Survivorship	Acupuncture	Sham acupuncture at placebo points
Lu 2016[77]	USA	42	Head and neck cancer	AOT	Electroacupuncture	Sham acupuncture at placebo points
Lu 2020[76]	USA	40	Breast cancer	Survivorship	Electroacupuncture	TAU
Mao 2014[79]	USA	67	Breast cancer	Survivorship	Electroacupuncture	Sham acupuncture at placebo points
Mao 2015[78]	USA	120	Breast cancer	Survivorship	Electroacupuncture	Sham acupuncture at placebo points
McKeon 2015[80]	Australia	60	Diverse cancer	AOT	Electroacupuncture	Sham acupuncture at placebo points
Meng 2012[82a]	China	86	Nasopharyngeal carcinoma	AOT	Acupuncture	TAU

(Continued)

TABLE 3. *Continued*

Reference	Origin	Sample size	Condition	Treatment Phase	Intervention (Type of Acupuncture)	Control Intervention(s)
Meng 2012[81]	China	23	Nasopharyngeal carcinoma	AOT	Acupuncture	Sham acupuncture at placebo points
Minchom 2016[83]	UK	173	Lung cancer	AOT (palliative)	1) Acupuncture alone; 2) acupuncture and morphine	TAU (morphine)
Molassiotis 2007[86]	UK	47	Diverse cancer	AOT	Acupuncture	Active control (acupressure)
Molassiotis 2012[84]	UK	302	Breast cancer	Survivorship	Acupuncture	TAU
Molassiotis 2019[85]	Hong Kong	87	Diverse cancer	AOT	Acupuncture	TAU
Nedstrand 2005[87]	Sweden	38	Breast cancer	Survivorship	Electroacupuncture	Active control (applied relaxation)
Ng 2013[88]	Hong Kong	165	Colorectal cancer	Surgery	Electroacupuncture	Sham acupuncture at placebo points
Ntritsou 2014[89]	Greece	75	Prostate cancer	Surgery	Electroacupuncture	Sham acupuncture at placebo points
Oh 2013[90]	Australia and USA	32	Breast cancer	Survivorship	Electroacupuncture	Sham acupuncture at acupuncture points
Pfister 2010[91]	USA	70 (58 analyzed)	Neck cancer	AOT	Acupuncture	TAU
Rithirangsriroj 2015[92a]	Thailand	70	Gynecologic cancer	AOT	Acupuncture (wrist)	TAU
Rostock 2013[93]	Germany, Switzerland	60	Diverse cancer	Survivorship	Electroacupuncture	Active control (hydroelectric baths)
Shen 2000[94]	USA	104	Breast cancer	AOT	Electroacupuncture	Sham electroacupuncture at placebo points
Simcock 2013[95]	UK	145	Head and neck cancer	Survivorship	Acupuncture	TAU
Smith 2013[96]	Australia	30	Breast cancer	Survivorship	Acupuncture	Sham acupuncture at placebo points
Streitberger 2003[97]	Germany	80	Diverse cancer	AOT	Acupuncture	Sham acupuncture at acupuncture points
Tong 2018[98]	China	80	Breast cancer	AOT	Acupuncture	TAU
Vickers 2005[99]	USA	47	Lung or breast cancer	AOT (palliative care)	Acupuncture followed by acupressure	Sham acupuncture at placebo points followed by placebo acupressure
Walker 2009[100]	USA	50	Breast cancer	Survivorship (100% antiestrogen hormone treatment)	Acupuncture	Active control (venlafaxin)
Wang & Wang. 2019[101]	China	60	Diverse cancer	Survivorship	Acupuncture	TAU
Wong 2006[102]	China	27	Nonsmall cell lung carcinoma	Surgery	Electroacupuncture	Sham acupuncture at acupuncture points
Yang 2012[103]	China	80	Supratentorial tumors	Surgery	Electroacupuncture	Sham electroacupuncture at acupuncture points
Yang 2017[104]	China	170	Cervical cancer	Surgery	Acupuncture	TAU
Yu 2013[105]	China and Germany	58	Astrocytoma	Survivorship	Acupuncture	TAU
Zeng 2014[106]	China	60	Liver cancer	AOT	Acupuncture	TAU
Zhang 2014[107]	China	40[a]	Colon cancer	Surgery	Electroacupuncture	Sham acupuncture at placebo points

Abbreviations: AOT, active oncological treatment; CBT-I, cognitive behavioral therapy for insomnia; TAU, treatment as usual.
[a]These were crossover trials that had insufficient reporting for inclusion in the meta-analysis.

that 96% of the publications included AEs only if they were serious, 37% did not specify criteria for accessing AEs, and 88% grouped together various kinds of AEs.[139]

In the context of AEs, it also must be discussed whether these should be reported by clinicians and patients. In therapy for oncological patients and in acupuncture treatments, there seems to be a discrepancy between the assessment of AEs by physicians and patients.[140-142] In the future, clarifying research regarding the different assessments will be useful.

To our knowledge, this is the first meta-analysis to comprehensively address the safety of acupuncture in oncological patients. Some meta-analyses that focus on the effectiveness of acupuncture for treating symptoms in patients with breast cancer or pain in oncological patients do report AEs and conclude that acupuncture is a safe intervention.[25,31,32] A prospective, observational study on nononcological patients by Witt et al in 2009 indicated that acupuncture was a relatively safe treatment option[14]: common adverse effects were bleeding

Study	log[Odds Ratio]	SE	Weight	Odds Ratio: IV-Random [95% CI]
Arai 2013[44]	0	2.03809866	4.0%	1.00 [0.02-54.30]
Bokmand & Flyger 2013[49]	0.09237332	2.01535849	4.1%	1.10 [0.02-56.96]
Brinkhaus 2019[50]	4.51039178	1.43864438	8.1%	90.96 [5.42-1525.51]
Chae 2016[51]	0	2.0976177	3.8%	1.00 [0.02-61.02]
Crew 2007[54]	0	2.02366946	4.1%	1.00 [0.02-52.79]
D'Alessandro 2022[56]	-0.06899287	2.03423087	4.0%	0.93 [0.02-50.30]
Han 2017[67]	0	2.00959238	4.1%	1.00 [0.02-51.35]
Jeon 2019[69]	1.178655	1.68134562	5.9%	3.25 [0.12-87.71]
Jung 2017[71]	0	2.02484567	4.1%	1.00 [0.02-52.91]
Kong 2020[73]	-0.24116206	2.04018079	4.0%	0.79 [0.01-42.84]
Lu 2020[76]	1.14990558	1.66412086	6.0%	3.16 [0.12-82.39]
Meng 2012[81]	0.13976194	2.01165085	4.1%	1.15 [0.02-59.30]
Minchom 2016[83]	-1.28565084	1.56546746	6.8%	0.28 [0.01-5.95]
Molassiotis 2019[85]	-0.02298952	2.01146292	4.1%	0.98 [0.02-50.37]
Ng 2013[88]	0	2.00907034	4.1%	1.00 [0.02-51.30]
Shen 2000[94]	1.04145388	1.65041092	6.1%	2.83 [0.11-71.97]
Tong 2018[98]	0	2.01246118	4.1%	1.00 [0.02-51.64]
Wang & Wang 2019[101]	1.67843078	1.57132034	6.8%	5.36 [0.25-116.53]
Yang 2017[104]	2.38635498	1.49575219	7.5%	10.87 [0.58-203.97]
Yu 2013[105]	-0.66717169	2.01930628	4.1%	0.51 [0.01-26.86]
Total (95% CI)			**100.0%**	**2.09 [0.94-4.66]**

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 12.41$, df = 19 ($P = .87$); $I^2 = 0\%$
Test for overall effect: Z = 1.81 ($P = .07$)

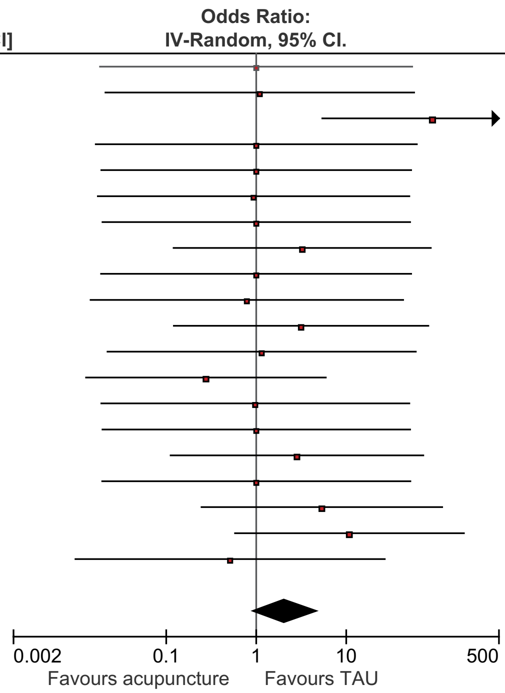

FIGURE 2. This forest plot compares acupuncture versus treatment as usual (TAU) for intervention-related adverse events. CI indicates confidence interval; df, degrees of freedom; IV-Random, random-effects models using the inverse-variance method; SE, standard error.

Study	log[Odds Ratio]	SE	Weight	Odds Ratio: IV-Random [95% CI]
Arai 2013[44]	0	2.03809866	9.7%	1.00 [0.02-54.30]
Bokmand & Flyger 2013[49]	0.09237332	2.01535849	9.9%	1.10 [0.02-56.96]
Brinkhaus 2019[50]	1.98673214	1.52083415	17.4%	7.29 [0.37-143.67]
Crew 2007[54]	0	2.02366946	9.8%	1.00 [0.02-52.79]
D'Alessandro 2022[56]	-0.07410797	1.46572564	18.7%	0.93 [0.05-16.42]
Lu 2020[76]	0	2.02484567	9.8%	1.00 [0.02-52.91]
Minchom 2016[83]	1.49392502	1.64844711	14.8%	4.45 [0.18-112.71]
Pfister 2010[91]	0.05715841	2.01424664	9.9%	1.06 [0.02-54.87]
Total (95% CI)			**100.0%**	**1.76 [0.51-6.11]**

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 1.73$, df = 7 ($P = .97$); $I^2 = 0\%$
Test for overall effect: Z = 0.89 ($P = .37$)

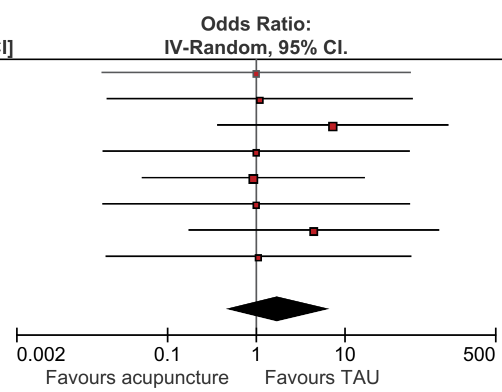

FIGURE 3. This forest plot compares acupuncture versus treatment as usual (TAU) for serious adverse events. CI indicates confidence interval; df, degrees of freedom; IV-Random, random-effects models using the inverse-variance method; SE, standard error.

or hematoma (6.1% of patients; 58% of all adverse effects), pain (1.7% of patients), and vegetative symptoms (0.7% of patients). In 229,230 patients, only 3 cases of pneumothorax were reported, and no acupuncture-associated deaths or permanent injuries were observed. In a current meta-analysis by Baumler et al of different patient populations, the authors found that an AE can be expected in every 10th patient or at every 13th acupuncture treatment.[15] Most of the AEs were minor and transient or even intended effects, indicating a favorable physiological response. Serious AEs were very rare. In our analysis, intervention-related AEs were observed in every 15th patient, and most AEs were nonserious. Acupuncture was not associated with an increased risk of serious AEs compared with sham acupuncture, TAU, or other active controls. Furthermore, there was no increased frequency of intervention-related AEs or drop-outs because of AEs in the acupuncture group compared

Study	log[Odds Ratio]	SE	Weight	Odds Ratio: IV- Random [95% CI]
Arai 2013[44]	0	2.03809866	2.1%	1.00 [0.02-54.30]
Bao 2018[47]	1.17272026	1.64806469	3.2%	3.23 [0.13-81.68]
Bokmand & Flyger 2013[49]	0.09237332	2.01535849	2.2%	1.10 [0.02-56.96]
Brinkhaus 2019[50]	1.11203531	1.64119271	3.3%	3.04 [0.12-75.85]
Chae 2016[51]	0	2.0976177	2.0%	1.00 [0.02-61.02]
Crew 2007[54]	0	2.02366946	2.1%	1.00 [0.02-52.79]
D'Alessandro 2022[56]	-0.07410797	1.46572564	4.1%	0.93 [0.05-16.42]
Garcia 2008[62]	0.05129329	2.01278806	2.2%	1.05 [0.02-54.40]
Giron 2016[64]	0	1.44337567	4.2%	1.00 [0.06-16.93]
Han 2017[67]	-1.11803038	1.64484202	3.2%	0.33 [0.01-8.21]
Jeon 2019[69]	1.178655	1.68134562	3.1%	3.25 [0.12-87.71]
Jung 2017[71]	0	2.02484567	2.1%	1.00 [0.02-52.91]
Kong 2020[73]	-0.24116206	2.04018079	2.1%	0.79 [0.01-42.84]
Mao 2014[79]	0.04445176	2.02211098	2.1%	1.05 [0.02-55.02]
McKeon 2015[80]	-0.04879016	2.02425765	2.1%	0.95 [0.02-50.33]
Meng 2012[81]	0.13976194	2.01165085	2.2%	1.15 [0.02-59.30]
Minchom 2016[83]	0.88551907	0.51468339	33.1%	2.42 [0.88-6.65]
Molassiotis 2019[85]	-0.02298952	2.01146292	2.2%	0.98 [0.02-50.37]
Ng 2013[88]	0	2.00907034	2.2%	1.00 [0.02-51.30]
Pfister 2010[91]	0.05715841	2.01424664	2.2%	1.06 [0.02-54.87]
Shen 2000[94]	-0.08455739	2.01406028	2.2%	0.92 [0.02-47.60]
Simcock 2013[95]	-1.05664809	1.64153922	3.2%	0.35 [0.01-8.68]
Smith 2013[96]	1.2039728	1.69640142	3.0%	3.33 [0.12-92.65]
Tong 2018[98]	0	2.01246118	2.2%	1.00 [0.02-51.64]
Yang 2017[104]	0	2.00890873	2.2%	1.00 [0.02-51.28]
Yu 2013[105]	0.45810884	1.65679683	3.2%	1.58 [0.06-40.67]
Zeng 2014[106]	0	2.0165978	2.2%	1.00 [0.02-52.06]
Total (95% CI)			**100.0%**	**1.46 [0.82-2.61]**

Heterogeneity: $Tau^2 = 0.00$; $\chi^2 = 4.25$, df = 26 ($P = 1.00$); $I^2 = 0\%$
Test for overall effect: $Z = 1.29$ ($P = 0.20$)

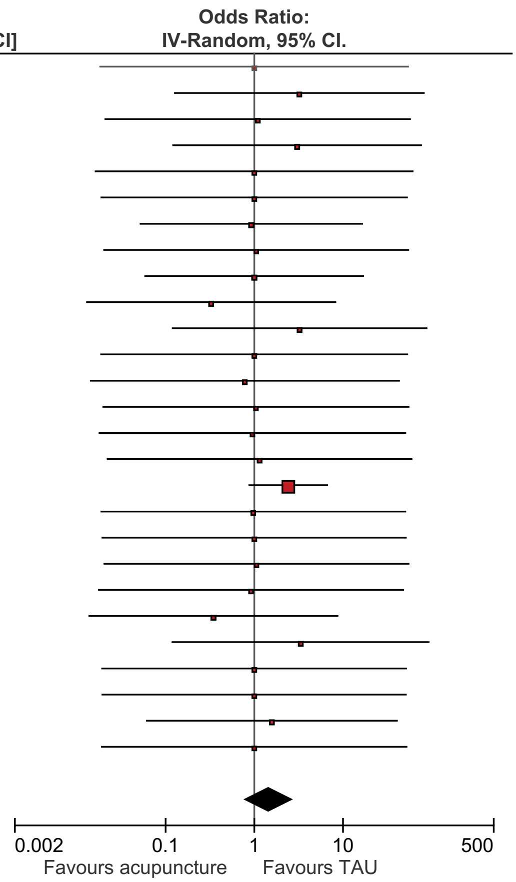

FIGURE 4. This forest plot compares acupuncture versus treatment as usual (TAU) for dropout because of adverse events. CI indicates confidence interval; df, degrees of freedom; IV-Random, random-effects models using the inverse-variance method; SE, standard error.

Study	log[Odds Ratio]	SE	Weight	Odds Ratio: IV- Random [95% CI]
Arai 2013[44]	0	2.03809866	8.0%	1.00 [0.02-54.30]
Bokmand & Flyger 2013[49]	1.84784894	1.1263945	20.0%	6.35 [0.70-57.72]
Brinkhaus 2019[50]	4.51039178	1.43864438	14.2%	90.96 [5.42-1525.51]
Crew 2007[54]	0	2.02366946	8.1%	1.00 [0.02-52.79]
D'Alessandro 2022[56]	-0.06899287	2.03423087	8.1%	0.93 [0.02-50.30]
Minchom 2016[83]	0.88551907	0.51468339	41.6%	2.42 [0.88-6.65]
Total (95% CI)			**100.0%**	**3.94 [1.16-13.35]**

Heterogeneity: $Tau^2 = 0.67$; $\chi^2 = 7.09$, df = 5 ($P = .21$); $I^2 = 29\%$
Test for overall effect: $Z = 2.20$ ($P = .03$)

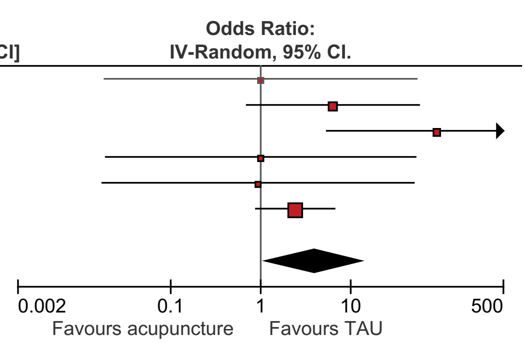

FIGURE 5. This forest plot compares acupuncture versus treatment as usual (TAU) for nonserious adverse events. CI indicates confidence interval; df, degrees of freedom; IV-Random, random-effects models using the inverse-variance method; SE, standard error.

with sham acupuncture, TAU, and other active controls. Only in the comparison between acupuncture and TAU was there an increased frequency of nonserious AEs, such as bruising or light pain. However, the included studies and the data obtained from them are not sufficient to classify acupuncture treatments as safe during phases of neutropenia and/or thrombocytopenia for patients on active oncological treatment. Although, in the pilot randomized, sham-controlled clinical trial by Lu et al of 21 patients who had ovarian cancer, acupuncture was administered during myelosuppressive chemotherapy, and no serious AEs were observed.[128] The only intervention-related AE was mild bruising at needling sites. In addition, in a descriptive, retrospective case series of adult oncological patients with thrombocytopenia <100,00/μL in 98 individual acupuncture visits, including 9 visits with platelet counts <50,000/μL, no increased bleeding or bruising was observed.[143] Nevertheless, the data are not sufficient. Clinically, an increased risk of infection and bleeding should be considered and differential blood counts should be determined during immunosuppressive therapies to avoid acupuncture in severe neutropenic and thrombocytopenic situations.

There are several limitations of our meta-analysis. First, the insufficient reporting of the RCTs included in this meta-analysis is a distinct limitation. To mitigate this effect, we queried data from authors if the respective studies included at least some information on AEs but were not yet sufficient for inclusion in our analysis. Despite our efforts, only 69.1% of the studies reported sufficient data to be included in our meta-analysis. Because of contradictory reporting in many studies, the use of AE terminology seems to be challenging.[144-149] Furthermore, there was high heterogeneity in the included studies concerning study populations, study periods, treatment phases, interventions, and precision in reporting of AEs. Although some authors classified pain from needling as an AE, others reported no AEs while stating that there was pain from needling.

In future randomized trials, study authors should adhere to the CONSORT safety and harm extension for reporting to ensure that the safety of interventions can be accessed more adequately.[12] In particular, authors should report on repeated AEs as well as AEs in general, and not only on intervention-related AEs.

This meta-analysis suggests that, overall, the severity and frequency of AEs in RCTs in oncology are comparable between acupuncture and sham acupuncture and active control interventions. Concerning only nonserious AEs, acupuncture might be associated with more AEs compared with TAU. Regarding the frequency of serious AEs, we found no difference when comparing acupuncture with sham acupuncture, TAU, and active controls. Acupuncture has a low risk of AEs, comparable to that of sham acupuncture and active controls, in oncological patients. To date, data on the risk of AEs in oncological patients are consistent with the safety profile of acupuncture in the treatment of chronic pain.

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CONFLICT OF INTEREST DISCLOSURES

The authors made no disclosures.

AUTHOR CONTRIBUTIONS

Melanie D. Höxtermann: Conceptualization, writing, formal analysis, and methodology. **Heidemarie Haller:** Conceptualization, formal analysis, and methodology. **Shaimaa Aboudamaah:** Formal analysis. **Armin Bachemir:** Formal analysis. **Holger Cramer:** Conceptualization, formal analysis, and methodology. **Gustav Dobos:** Writing and funding acquisition. **Petra Voiss:** Conceptualization, writing, formal analysis, and funding acquisition.

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